

Autodesk® Scaleform®

Scaleform 3.3

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Autodesk® Scaleform® 4.4

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1 Scaleform 3.x 메모리 시스템

Autodesk® Scaleform® 3.0 (heap) ,
Scaleform 3.2
AMP(Analyzer for Memory and Performance)
Scaleform 3.3
1.2

Scaleform 3.3

-
- Movies Scaleform
- ActionScript (Garbage Collection)
-

1.1 중요한 메모리 개념

Scaleform / Scaleform

1.1.1 오버라이드 할당자

Scaleform Gfx::System

1. Scaleform::SysAlloc Alloc/Free/Realloc
dlmalloc

2. Scaleform

3. Scaleform (: Scaleform::SysAllocWinAPI)

(2)

2'

1.1.2 메모리 힙

Scaleform

AMP MemReport

- Global -
- MovieData - SWF/GFx
- MovieView - GFx::Movie ActionScript

- MeshCashe - shape triangle data

-
-
-

Scaleform 3.3

Scaleform::SysAlloc

, Scaleform 3.3

GfX::Movie

1.1.3 조각모음

Scaleform 3.0 AS(ActionScript)

AS

CLIK

ActionScript

Scaleform GfX::Movie

, AS

. Scaleform 3.3 Movie

(garbage collector)

ActionScript

Movie

Scaleform

4:

1.1.4 메모리 보고 및 디버깅

Scaleform "stat ID"

stat ID

stat ID

Scaleform 3.2

AMP(Analyzer for Memory and Performance)

. AMP

[AMP](#)

, Scaleform::MemoryHeap::MemReport

stat ID

. Scaleform

Visual Studio

. Scaleform

Scaleform

Shipping

1.2 Scaleform 3.3 메모리 API 변경점

Scaleform 3.3

- Gfx::Movie

movie view

4

- SysAlloc

"malloc

"

4K

512

SysAlloc

Scaleform::SysAlloc

Scaleform::SysAllocPaged

. SysAlloc

SysAllocPaged

. SysAllocStatic

2 메모리 할당

1.1.1 , Scaleform

.

- 1. SysAlloc
- 2. Scaleform
- 3. Scaleform

” ’ . E E ∞ †

WF K

K

ñ

2.1 할당 오버라이딩

Scaleform

1) SysAlloc . Alloc, Free, Realloc

2) Scaleform Gfx::System

malloc/free

Scaleform::SysAllocMalloc Scaleform

2.1.1 자신만의 SysAlloc 구현

Scaleform SysAllocMalloc

SysAllocMalloc

```
class MySysAlloc : public SysAlloc
{
public:
    virtual void* Alloc(UPInt size, UPInt align)
    {
        return _aligned_malloc(size, align);
    }

    virtual void Free(void* ptr, UPInt size, UPInt align)
    {
        SF_UNUSED2(size, align);
        _aligned_free(ptr);
        return true;
    }

    virtual void* Realloc(void* oldPtr, UPInt oldSize,
                          UPInt newSize, UPInt align)
    {
        SF_UNUSED(oldSize);
        return _aligned_realloc(oldPtr, newSize, align);
    }
};
```

```

        . MySysAlloc      SysAlloc
3      ( Alloc, Free, Realloc).
      , Scaleform      16      . oldSize      align
      Free/Realloc      .      , Alloc/Free
      wrapper      Realloc      .
      ,      Scaleform
Gfx::System
      .
      MySysAlloc myAlloc;
      GfxSystem gfxSystem(&myAlloc);

Scaleform 3.0      Gfx::System      Scaleform      ,
Scaleform      . Scaleform
      .      , Scaleform
      (Gfx::System      않는      ).
      Gfx::System::Init()      Gfx::System::Destroy()
      . Gfx::System      Gfx::System::Init()
SysAlloc      .

```

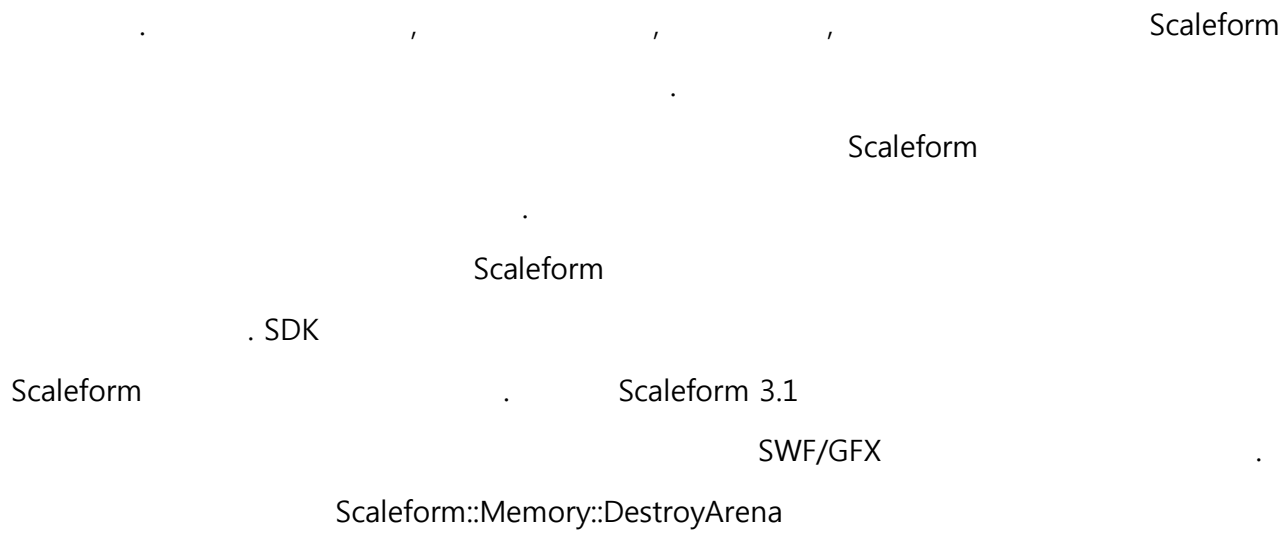
2.2 Scaleform 에 고정 메모리 블록 사용하기

```

      1
      , SysAlloc      Scaleform      .
      SysAllocStatic      .

void*      pmemChunk = ...;
SysAllocStatic      blockAlloc(pmemChunk, 6*1024*1024);
System      gfxSystem(&blockAlloc);
...

```

2.2.1.2 메모리 아레나의 사용

1. Scaleform::Memory::CreateArena 0(0 |
)
SysAlloc SysAllocStatic

// Assume pbuf1 points to a buffer of 10,000,000 bytes.
SysAllocStatic sysAlloc(pbuf1, 10000000);
Memory::CreateArena(1, &sysAlloc);
2. ID Gfx::Loader::CreateMovie/Gfx::MovieDef::CreateInstance
movie
“ ”

// use arena 1 for the movie
Ptr<MovieDef> pMovieDef =

```

        *Loader.CreateMovie(filenameStr, Loader::LoadWaitFrame1, 1);
...
// use arena 1 for the instance
Ptr<Movie> pMovie =
    *pnewMovieDef->CreateInstance(MovieDef::MemoryParams(1), false);

3.          Gfx::MovieDef/Gfx::Movie
4.          . Gfx::MovieDef::WaitForLoadFinish
          . Gfx::ThreadedTaskManager

pMovieDef->WaitForLoadFinish(true);
pMovie = 0;
pMovieDef = 0;

( Scaleform::Ptr ):
pMovie->Release();
pMovieDef->Release();

5. DestroyArena . CreateArena
          . Scaleform::Memory::ArenaIsEmpty

// DestroyArena will ASSERT if memory was not properly released before
// the call. In Release it will crash at "return *(int*)0;".
Memory::DestroyArena(1);

SysAllocobject 'pbuff1' (scope)

```

2.3 OS

- Scaleform OS
- *Scaleform::SysAllocWinAPI* – *VirtualAlloc()/VirtualFree()* API . MS

- *Scaleform::SysAllocPS3* –PS3 `sys_memory_allocate/sys_memory_free`

OS

Scaleform WinAPI PS3 SysAlloc 64K

`dlmalloc`

2MB

2.3.1 SysAllocPaged 구현하기

Scaleform 3.3 SysAlloc , SysAlloc

SysAllocPaged . SysAllocPaged

Scaleform , SysAllocStatic OS

SysAlloc SysAllocPaged

SysAllocPagedMalloc

```
class MySysAlloc : public SysAllocPaged
{
public:
    virtual void GetInfo(Info* i) const
    {
        i->MinAlign    = 1;
        i->MaxAlign    = 1;
        i->Granularity  = 128*1024;
        i->HasRealloc   = false;
    }

    virtual void* Alloc(UPInt size, UPInt align)
    {
        // Ignore 'align' since reported MaxAlign is 1.
        return malloc(size);
    }
}
```



```

    }

    virtual bool Free(void* ptr, UInt size, UInt align)
    {
        // free() doesn't need size or alignment of the memory block, but
        // you can use it in your implementation if it makes things easier.
        free(ptr);
        return true;
    }
};

```

```

        . MySysAlloc      SysAllocPaged
        3                (GetInfo, Alloc, Free).      ReallocInPlace
    ,
    .

```

- GetInfo() –Scaleform::SysAllocPaged::Info

- Alloc –


```

                .
                GetInfo  MaxAlign      . MaxAlign      1
            
```

- Free –Alloc


```

                . free
            
```

- ReallocInPlace –


```

                .
                false
            
```

[Scaleform Reference Documentation](#)

```

        Alloc  Free
GetInfo      .
        SysAllocPaged      .
        . SysAllocPaged::Info      4
    .

```

- MinAlign –

- `MaxAlign` – `0` `Scaleform`
`1` `Alloc`
- `Granularity` – `Scaleform` `Scaleform`
`( malloc )` `64K`
- `HasRealloc` – `ReallocInPlace` `false`
- `SysDirectThreshold` - `null`
`SysDirectThreshold` ,
- `MaxHeapGranularity` - `null` , `MaxHeapGranularity`
`alloc/free`
`MaxHeapGranularity`
`MaxHeapGranularity 4096 , 4096`

`MaxAlign` `SysAlloc`

3

1. `MaxAlign 1` `Scaleform`
2. `MaxAlign 0`
`Alloc`
3. `MinAlign` `MaxAlign`
`OS`

Scaleform

GfX::System

```
MySysAlloc myAlloc;  
GfX::System gfxSystem(&myAlloc);
```

2.4

Scaleform

mesh

cache

Scaleform

API

2.4.1 힙 API

Scaleform

'new'

```
Ptr<FileOpener> opener = *new FileOpener();  
loader.SetFileOpener(opener);
```

'new'

Scaleform

(global)

GfX::System

0

GfX::MeshCacheManager

shape

. GfX::Loader::CreateMovie()

SWF/GFX

. GfX::MovieDef::CreateInstance


```

        Gfx::SpriteDef      pHeap
    .

    Ptr<SpriteDef> def = *SF_HEAP_NEW(pHeap) SpriteDef(pdataDef);

    Gfx::SpriteDef      Scaleform      .      Scaleform
        Ptr<>
            . Scaleform::NewOverrideBase
        'delete'      . delete      SF_FREE(address)
    .

```

2.4.2 자동 힙

```

    .
        Scaleform::String/Scaleform::Array<>
    .

```

```

class Sprite : public RefCountBase<Sprite>
{
    String      Name;
    Array<Ptr<Sprite> > Children;

    Sprite(String& name, ...) { ... }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
sprite->DoSomething();

```

```

Gfx::Sprite      ,
    .
    ,
    .
        Gfx::Sprite      ,
    .
    ,

```

- Scaleform
- SF_HEAP_AUTO_ALLOC(ptr, size)
 - SF_HEAP_AUTO_NEW(ptr) ClassName

SF_HEAP_ALLOC SF_HEAP_NEW

```
MemoryHeap* pheap = ...;
UByte * pbuffer = (UByte*)SF_HEAP_ALLOC(pheap, 100,
StatMV_ActionScript_Mem);
Ptr<Sprite> sprite = *SF_HEAP_AUTO_NEW(pbuffer + 5) Sprite();
...
sprite->DoSomething();
...
// Release sprite and free buffer memory.
sprite = 0;
SF_FREE(pbuffer);
```

Gfx::Sprite

'pbuffer' SF_HEAP_AUTO_NEW

, Scaleform

Gfx::Sprite

```
class Sprite : public RefCountBase<Sprite>
{
    StringLH Name;
    ArrayLH<Ptr<Sprite> > Children;

    Sprite(String& name, ...) { ... }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
```

```

        sprite->DoSomething();
        Scaleform::String  Scaleform::Array<>          "          (Local Heap)"
Scaleform::StringLH  Scaleform::ArrayLH<>          .
        "          "          .
        .          Scaleform
        .

```

2.4.3 힙 추적기(Heap Tracer)

```

GFxConfig.h          SF_MEMORY_TRACE_ALL
Scaleform::Memory          SetTracer          CreateHeap, DestroyHeap, Alloc, Free
        .
        SysAllocWinAPI
        .

```

```

#include "Kernel/HeapPT/HeapPT_SysAllocWinAPI.h"
class MyTracer : public Scaleform::MemoryHeap::HeapTracer
{
public:
    virtual void OnCreateHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnDestroyHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnAlloc(const MemoryHeap* heap, UPInt size, UPInt align,
                        unsigned sid, const void* ptr)
    {
        //...
    }

    virtual void OnRealloc(const MemoryHeap* heap, const void* oldPtr,
                        UPInt newSize, const void* newPtr)
    {
        //...
    }
}

```

```

        virtual void OnFree(const MemoryHeap* heap, const void* ptr)
        {
            //...
        }
};

static MyTracer tracer;
static SysAllocWinAPI sysAlloc;
static GFx::System* gfxSystem;

class MySystem : public Scaleform::System
{
public:
    MySystem() : Scaleform::System(&sysAlloc)
    {
        Scaleform::Memory::GetGlobalHeap()->SetTracer(&tracer);
    }
    ~MySystem() { }
};

SF_PLATFORM_SYSTEM_APP(FxPlayer, MySystem, FxPlayerApp)

```


3 메모리 보고

Scaleform Player AMP HUD

Scaleform 3.0

PC

Scaleform

AMP

AMP

AMP

Scaleform Player HUD UI , (Ctrl + F5)

. Scaleform::MemoryHeap::MemReport

```
void MemReport (class StringBuffer& buffer, MemReportType detailed,
                bool xmlFormat = false);
void MemReport (struct MemItem* rootItem, MemReportType detailed);
```

MemReport, XML

MemReportType

- `MemoryHeap::MemReportBrief` – Scaleform

Memory 4,680K / 4,737K

Image	1,052
Sound	136
Movie View	1,739,784
Movie Data	300,156

- `MemoryHeap::MemReportFull` – Ctrl+F6

SF_MEMORY_ENABLE_DEBUG_INFO	stat ID	memory
breakdown	.	.

Memory 4,651K / 4,707K

System Summary

System Memory FootPrint	4,832,440
System Memory Used Space	4,775,684
Debug Heaps Footprint	12,884
Debug Heaps Used Space	5,392

Summary

Image	1,052
Sound	136
Movie View	1,715,128
Movie Data	300,156

[Heap] Global 4,848,864

Heap Summary

Total Footprint	4,848,864
Local Footprint	2,427,860
Child Footprint	2,421,004
Child Heaps	4
Local Used Space	2,417,196
DebugInfo	120,832

Memory

MovieDef

Sounds	8
ActionOps	80,476
MD_Other	200

MovieDef

36

MovieView

MV_Other	32
----------	----

General

91,486

Image

92

Sound

136

String

31,425

Debug Memory

StatBag	16,384
---------	--------

Renderer

Buffers	2,097,152
RenderBatch	5,696
Primitive	1,512
Fill	900
Mesh	4,884
MeshBatch	2,200
Context	15,184
NodeData	15,160
TreeCache	13,220
TextureManager	2,336
MatrixPool	4,096
MatrixPoolHandles	2,032
Text	1,232

Renderer	13,488
Allocations Count	1,822

- MemoryHeap:: MemReportFileSummary –

.	SWF	MovieData, MovieDef	MovieViews	stat ID
.	.	.	.	.
Movie File	3DGenerator_AS3.swf			
Memory			1,956,832	
MovieDef			152,664	
CharDefs			39,848	
ShapeData			1,716	
Tags			73,656	
MD_Other			37,444	
MovieView			196,249	
MovieClip			120	
ActionScript			181,917	
MV_Other			14,212	
General			1,596,712	
Image			960	
String			2,783	
Renderer			7,464	
Text			7,464	

- MemoryHeap::MemReportHeapDetailed – AMP

Scaleform . :

Total Footprint	4,858,148
Used Space	4,793,972
Global Heap	2,405,492
MovieDef	80,720
Sounds	8
ActionOps	80,476
MD_Other	200
MovieView	32
MV_Other	32
General	92,566
Image	92
Sound	136
String	31,276
Debug Memory	8,192
StatBag	8,192
Renderer	2,182,960

Buffers	2,097,152
RenderBatch	5,696
Primitive	1,512
Fill	900
Mesh	4,884
MeshBatch	2,200
Context	15,184
NodeData	15,016
TreeCache	13,136
TextureManager	2,336
MatrixPool	8,192
MatrixPoolHandles	2,032
Text	1,232
Movie Data Heaps	300,156
MovieData "3DGenerator_AS3.swf"	300,156
MovieDef	152,664
CharDefs	39,848
ShapeData	1,716
Tags	73,656
MD_Other	37,444
General	141,332
Image	960
String	2,312
Movie View Heaps	1,727,936
MovieView "3DGenerator_AS3.swf"	1,727,936
MovieView	195,977
MovieClip	120
ActionScript	181,645
MV_Other	14,212
General	1,467,668
String	471
Renderer	7,464
Text	7,464
Other Heaps	360,580
_FMOD_Heap	360,580
General	359,944
Debug Data	12,884
Unused Space	51,292
Global	51,292
_FMOD_Heap	4,384
MovieData "3DGenerator_AS3.swf"	5,356
MovieView "3DGenerator_AS3.swf"	33,032

- MovieDef –
- MovieView – GfX::Movie
movie clip
ActionScript
- CharDefs – movie clip flash
- ShapeData –
- Tags – keyframe
- ActionOps – ActionScript
- MeshCache – tessellator EdgeAA
- FontCache –

4 가베지 콜렉션

Scaleform 2.0

2 (circular references)

Code:

```
var o1 = new Object;  
var o2 = new Object;  
o1.a = o2;  
o2.a = o1;
```

Scaleform 3.0

Scaleform 3.0

Scaleform

Include/GFxConfig.h

```
GFX_AS_ENABLE_GC  
// Enable garbage collection  
#define GFX_AS_ENABLE_GC
```

Scaleform

4.1

Scaleform 3.0

. Scaleform 3.3

, MovieView

MovieDef

```
Movie* MovieDef::CreateInstance(const MemoryParams& memParams,
                                bool initFirstFrame = true)
```

MemoryParams

```
struct MemoryParams
{
    MemoryHeap::HeapDesc    Desc;
    float                   HeapLimitMultiplier;
    unsigned                 MaxCollectionRoots;
    unsigned                 FramesBetweenCollections;
};
```

- Desc – (MinAlign), (Granularity), (Limit), (Flags) (Limit) , Scaleform
- HeapLimitMultiplier – (0~1).
- MaxCollectionRoots – roots , root
- FramesBetweenCollections – roots root

movie view

```
MemoryContext* MovieDef::CreateMemoryContext(
    const char* heapName,
    const MemoryParams& memParams,
```

```

        bool debugHeap)

Movie* MovieDef::CreateInstance(MemoryContext* memContext,
                                bool initFirstFrame = true)

MemoryContext      movie view
    . movie view      movie view
    - (thread-safe)    AMP
    AMP
        movie view
    .
    .

MemoryParams::Desc
    . ( :ActionScript ) (Desc.Limit, HeapLimitMultiplier)
    . 128K .(
    .)
    ,
    . Boehm-Demers-Weiser(BDW)

```



```

        + Delta
    (
        ).

    MaxCollectionRoots/ FramesBetweenCollections.MaxCollectionRoots
        root . "root"
        ( ) root . ( : "obj.member=1" "obj" .)
    MaxCollectionRoots
        0
    FramesBetweenCollections
    "frame"
    FramesBetweenCollections 1800 30fps
    60 . DescLimit
    FramesBetweenCollections
    0 ,

```

4.2 Gfx::Movie:: ForceCollectGarbage

```

virtual void ForceCollectGarbage() = 0;

```